

Fundamentals of Information Systems Security

Wireshark Display Filters Cheat Sheet

Version: 1.0

The World’s Most Popular Network Protocol Analyzer

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1 Filter Syntax Basics

Filter Syntax Rules

- **Protocol:** Filter by protocol name (e.g., tcp, udp, http, dns)
- **Field:** Filter by field (e.g., ip.src, tcp.port)
- **Comparison:** == != > < >= <=
- **Logical:** and, or, xor, not (and &&, ||, !)
- **Contains:** contains, matches
- **Parentheses:** Group for complex filters

2 Protocol Filters

Filter	Description	Example
tcp	All TCP traffic	tcp
udp	All UDP traffic	udp
icmp	All ICMP traffic	icmp
http	All HTTP traffic	http
https	All TLS/SSL (HTTPS)	tls or ssl
dns	All DNS traffic	dns
arp	All ARP traffic	arp
ftp	All FTP traffic	ftp
ssh	All SSH traffic	ssh
telnet	All Telnet traffic	telnet
smb	All SMB traffic	smb
snmp	All SNMP traffic	snmp
dhcp	All DHCP traffic	bootp
ntp	All NTP traffic	ntp

3 IP Address Filters

Filter	Description	Example
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ip.addr == X.X.X.X	Traffic to/from IP	ip.addr == 192.168.1.1
ip.src == X.X.X.X	Source IP	ip.src == 192.168.1.1
ip.dst == X.X.X.X	Destination IP	ip.dst == 192.168.1.1
ip.addr != X.X.X.X	Not to/from IP	ip.addr != 192.168.1.1
ip.addr in {a,b,c}	IP in list	ip.addr in {10.0.0.1, 10.0.0.2}
ip.addr == 192.168.0.0/16	Subnet	ip.addr == 192.168.0.0/16

4 Port Filters

Filter	Description	Example
tcp.port == 80	TCP port 80 (HTTP)	tcp.port == 80
udp.port == 53	UDP port 53 (DNS)	udp.port == 53
tcp.port in {80,443}	Multiple TCP ports	tcp.port in {80,443}
tcp.srcport == 80	Source port	tcp.srcport == 80
tcp.dstport == 80	Destination port	tcp.dstport == 80
tcp.port >= 1024	Port range	tcp.port >= 1024

5 HTTP Filters

Filter	Description	Example
http	All HTTP traffic	http
http.request	HTTP requests	http.request
http.response	HTTP responses	http.response
http.request.method == "GET"	GET requests	http.request.method == "GET"
http.request.method == "POST"	POST requests	http.request.method == "POST"
http.response.code == 200	Status 200 OK	http.response.code == 200
http.response.code >= 400	Error responses	http.response.code >= 400
http contains "password"	Contains string	http contains "password"
http.request.uri contains "login"	URI contains	http.request.uri contains "login"
http.cookie	HTTP cookies	http.cookie

6 DNS Filters

Filter	Description	Example
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dns	All DNS traffic	dns
dns.qry.name	Query name	dns.qry.name contains "google"
dns.qry.type == 1	A record queries	dns.qry.type == 1
dns.qry.type == 28	AAAA record queries	dns.qry.type == 28
dns.flags.response == 0	DNS queries	dns.flags.response == 0
dns.flags.response == 1	DNS responses	dns.flags.response == 1
dns.resp.name	Response name	dns.resp.name == "example.com"

7 SSL/TLS Filters

Filter	Description	Example
tls	TLS traffic	tls
ssl	SSL traffic	ssl
tls.handshake.type == 1	Client Hello	tls.handshake.type == 1
tls.handshake.type == 2	Server Hello	tls.handshake.type == 2
tls.handshake.certificate	Certificates	tls.handshake.certificate
tls.handshake.extensions_server_name	SNI (Server Name Indication)	tls.handshake.extensions_server_name
tls.alert.message	TLS alerts	tls.alert.message

8 ARP Filters

Filter	Description	Example
arp	All ARP traffic	arp
arp.opcode == 1	ARP request	arp.opcode == 1
arp.opcode == 2	ARP reply	arp.opcode == 2
arp.src.proto_ipv4	Sender IP	arp.src.proto_ipv4 == 192.168.1.1
arp.dst.proto_ipv4	Target IP	arp.dst.proto_ipv4 == 192.168.1.1

9 Advanced Filters

Filter	Description
tcp.analysis.flags	TCP analysis flags
tcp.analysis.retransmission	TCP retransmissions
tcp.analysis.ack_lost	Missing ACKs

tcp.analysis.duplicate_ack	Duplicate ACKs
tcp.analysis.zero_window	Zero window notification
tcp.analysis.zero_window_probe	Zero window probe
frame.time_relative > 1	Packets after 1 second
frame.len > 1500	Large frames (jumbo)
data.data contains "secret"	Data contains string

10 Capture Filters (BPF)

Berkeley Packet Filters (BPF)

```
# Host filters
host 192.168.1.1
src host 192.168.1.1
dst host 192.168.1.1

# Port filters
port 80
src port 80
dst port 80
portrange 80-443

# Protocol filters
tcp
udp
icmp

# Network filters
net 192.168.0.0/24

# Complex filters
host 192.168.1.1 and port 80
tcp and port 443
host 192.168.1.1 or host 192.168.1.2
not arp

# Examples
tcp port 80 and host 192.168.1.1
icmp and host 192.168.1.1
udp port 53 and host 192.168.1.1
```

11 Common Filter Examples

Real-World Filter Examples

```
# HTTP GET requests to specific domain
http.request and http.host contains "example.com"

# HTTP POST with credentials
http.request.method == "POST" and http contains "password"

# DNS queries for suspicious domains
dns.qry.name contains "malicious" or dns.qry.name contains "phishing"

# TCP SYN packets (new connections)
tcp.flags.syn == 1 and tcp.flags.ack == 0

# Suspicious large data transfers
tcp.len > 10000

# ARP spoofing detection (multiple MACs for same IP)
arp.duplicate-address-frame

# TLS handshake with SNI
tls.handshake.type == 1 and tls.handshake.extensions_server_name
```